CSE 3204
Formal Language, Automata and Computability

Lecture 7
Finite Automata, Regular Expression and Regular Languages

Converting Regular Expression to Automata (∈-NFA)

Theorem 3.7: Every language defined by a regular expression is also defined by a finite automaton.

PROOF: Suppose $L = L(R)$ for a regular expression R. We show that $L = L(E)$ for some ϵ-NFA E with:

1. Exactly one accepting state.
2. No arcs into the initial state.
3. No arcs out of the accepting state.

The proof is by structural induction on R, following the recursive definition of regular expressions that we had in Section 3.1.2.

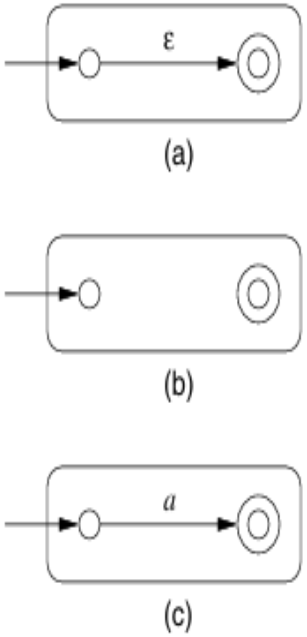


Figure 3.16: The basis of the construction of an automaton from a regular expression

Converting Regular Expression to Automata (∈-NFA)

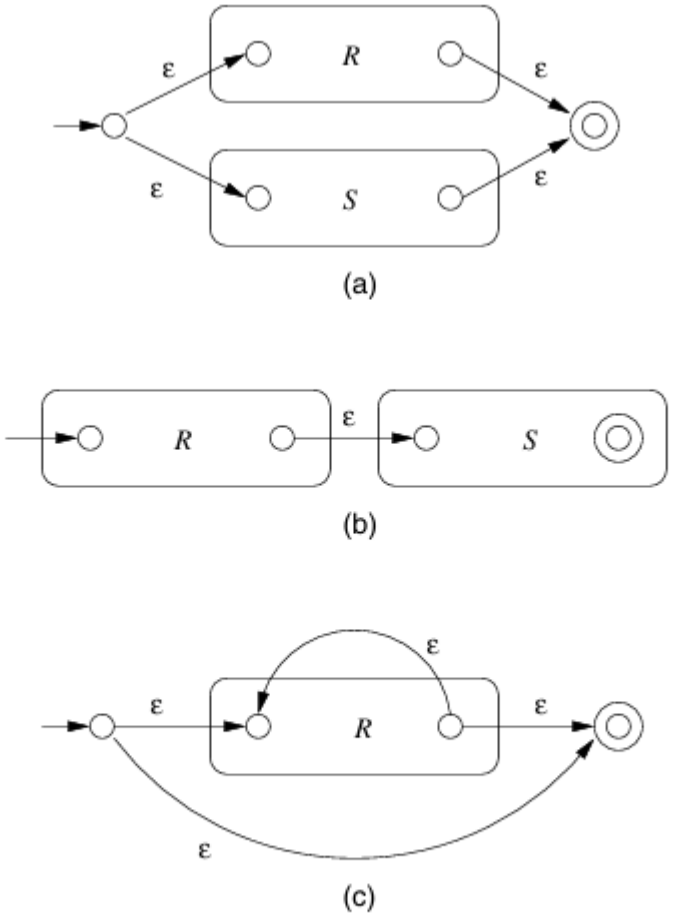


Figure 3.17: The inductive step in the regular-expression-to-ϵ-NFA construction

Conversion of $(0+1)^*1(0+1)$ to an ∈-NFA

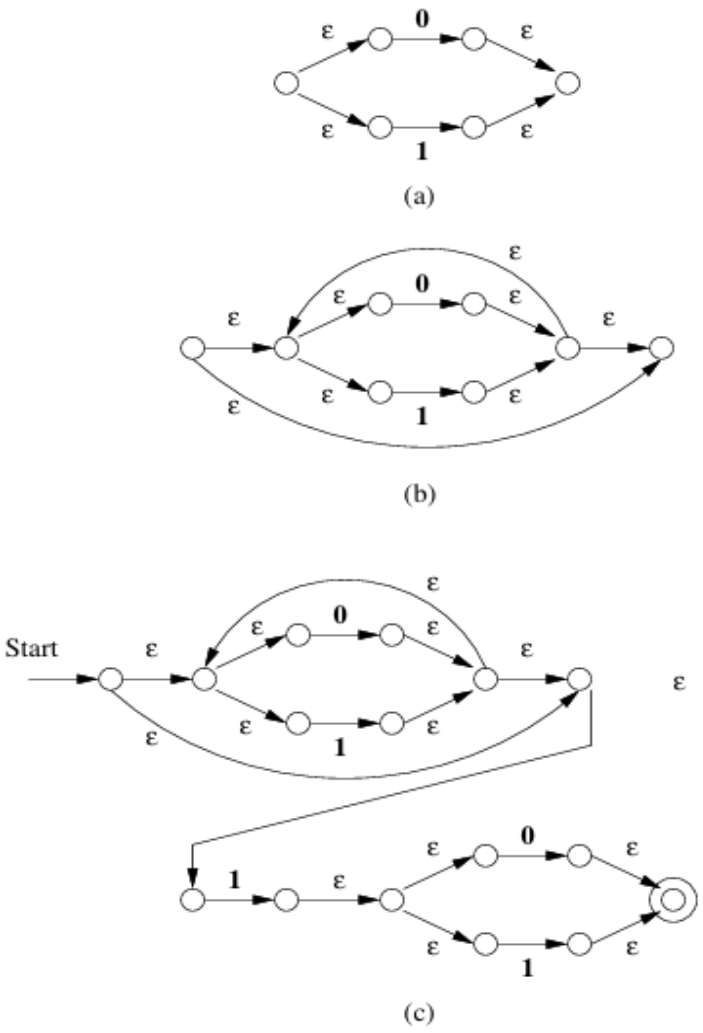
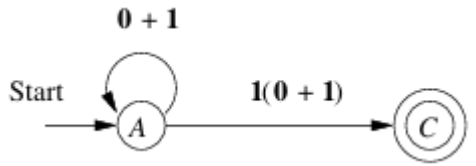


Figure 3.18: Automata constructed for Example 3.8



Equivalent NFA

Algebraic Laws for Regular Expression Associativity and Commutativity

- $L + M = M + L$. This law, the *commutative law for union*, says that we may take the union of two languages in either order.
- $(L + M) + N = L + (M + N)$. This law, the *associative law for union*, says that we may take the union of three languages either by taking the union of the first two initially, or taking the union of the last two initially.
- $(LM)N = L(MN)$. This law, the *associative law for concatenation*, says that we can concatenate three languages by concatenating either the first two or the last two initially.

Commutativity law for concatenation does not hold for regular expressions.

Identities and Annihilators

There are three laws for regular expressions involving these concepts; we list them below.

- $\emptyset + L = L + \emptyset = L$. This law asserts that $\emptyset$ is the identity for union.
- $\epsilon L = L\epsilon = L$. This law asserts that ϵ is the identity for concatenation.
- $\emptyset L = L\emptyset = \emptyset$. This law asserts that $\emptyset$ is the annihilator for concatenation.

No annihilator for union.

Distributive Laws

- $L(M + N) = LM + LN$. This law, is the *left distributive law of concatenation over union.*
- $(M + N)L = ML + NL$. This law, is the *right distributive law of concatenation over union.*

Left Distributive Law of Concatenation

Theorem 3.11: If L, M, and N are any languages, then

$$L(M \cup N) = LM \cup LN$$

PROOF: The proof is similar to another proof about a distributive law that we saw in Theorem 1.10. We need first to show that a string w is in $L(M \cup N)$ if and only if it is in $LM \cup LN$.

(Only-if) If w is in $L(M \cup N)$, then $w = xy$, where x is in L and y is in either M or N. If y is in M, then xy is in LM, and therefore in $LM \cup LN$. Likewise, if y is in N, then xy is in LN and therefore in $LM \cup LN$.

(If) Suppose w is in $LM \cup LN$. Then w is in either LM or in LN. Suppose first that w is in LM. Then $w = xy$, where x is in L and y is in M. As y is in M, it is also in $M \cup N$. Thus, xy is in $L(M \cup N)$. If w is not in LM, then it is surely in LN, and a similar argument shows it is in $L(M \cup N)$. □

Example 3.12: Consider the regular expression $\mathbf{0}+\mathbf{01}^*$. We can "factor out a $\mathbf{0}$" from the union, but first we have to recognize that the expression $\mathbf{0}$ by itself is actually the concatenation of $\mathbf{0}$ with something, namely ϵ. That is, we use the identity law for concatenation to replace $\mathbf{0}$ by $\mathbf{0}\epsilon$, giving us the expression $\mathbf{0}\epsilon+\mathbf{01}^*$. Now, we can apply the left distributive law to replace this expression by $\mathbf{0}(\epsilon+\mathbf{1}^*)$. If we further recognize that ϵ is in $L(\mathbf{1}^*)$, then we observe that $\epsilon+\mathbf{1}^*=\mathbf{1}^*$, and can simplify to $\mathbf{01}^*$. □

The Idempotent Law

- $L + L = L$. This law, the *idempotence law for union*, states that if we take the union of two identical expressions, we can replace them by one copy of the expression.

3.4.5 Laws Involving Closures

There are a number of laws involving the closure operators and its UNIX-style variants $^+$ and ?. We shall list them here, and give some explanation for why they are true.

- $(L^*)^* = L^*$. This law says that closing an expression that is already closed does not change the language. The language of $(L^*)^*$ is all strings created by concatenating strings in the language of L^*. But those strings are themselves composed of strings from L. Thus, the string in $(L^*)^*$ is also a concatenation of strings from L and is therefore in the language of L^*.
- $\emptyset^* = \epsilon$. The closure of $\emptyset$ contains only the string ϵ, as we discussed in Example 3.6.
- $\epsilon^* = \epsilon$. It is easy to check that the only string that can be formed by concatenating any number of copies of the empty string is the empty string itself.
- $L^+ = LL^* = L^*L$. Recall that L^+ is defined to be $L + LL + LLL + \cdots$. Also, $L^* = \epsilon + L + LL + LLL + \cdots$. Thus,

  $$LL^* = L\epsilon + LL + LLL + LLLL + \cdots$$

  When we remember that $L\epsilon = L$, we see that the infinite expansions for LL^* and for L^+ are the same. That proves $L^+ = LL^*$. The proof that $L^+ = L^*L$ is similar.[4]
- $L^* = L^+ + \epsilon$. The proof is easy, since the expansion of L^+ includes every term in the expansion of L^* except ϵ. Note that if the language L contains the string ϵ, then the additional "$+\epsilon$" term is not needed; that is, $L^+ = L^*$ in this special case.
- $L? = \epsilon + L$. This rule is really the definition of the ? operator.

Pumping Lemma for Regular Languages

Theorem 4.1: (The *pumping lemma for regular languages*) Let L be a regular language. Then there exists a constant n (which depends on L) such that for every string w in L such that $|w| \geq n$, we can break w into three strings, $w = xyz$, such that:

1. $y \neq \epsilon$.
2. $|xy| \leq n$.
3. For all $k \geq 0$, the string $xy^k z$ is also in L.

That is, we can always find a nonempty string y not too far from the beginning of w that can be "pumped"; that is, repeating y any number of times, or deleting it (the case $k = 0$), keeps the resulting string in the language L.

PROOF: Suppose L is regular. Then $L = L(A)$ for some DFA A. Suppose A has n states. Now, consider any string w of length n or more, say $w = a_1 a_2 \cdots a_m$, where $m \geq n$ and each a_i is an input symbol. For $i = 0, 1, \ldots, n$ define state p_i to be $\hat{\delta}(q_0, a_1 a_2 \cdots a_i)$, where δ is the transition function of A, and q_0 is the start state of A. That is, p_i is the state A is in after reading the first i symbols of w. Note that $p_0 = q_0$.

By the pigeonhole principle, it is not possible for the $n+1$ different p_i's for $i = 0, 1, \ldots, n$ to be distinct, since there are only n different states. Thus, we can find two different integers i and j, with $0 \leq i < j \leq n$, such that $p_i = p_j$. Now, we can break $w = xyz$ as follows:

1. $x = a_1 a_2 \cdots a_i$.
2. $y = a_{i+1} a_{i+2} \cdots a_j$.
3. $z = a_{j+1} a_{j+2} \cdots a_m$.

That is, x takes us to p_i once; y takes us from p_i back to p_i (since p_i is also p_j), and z is the balance of w. The relationships among the strings and states are suggested by Fig. 4.1. Note that x may be empty, in the case that $i = 0$. Also, z may be empty if $j = n = m$. However, y can not be empty, since i is strictly less than j.

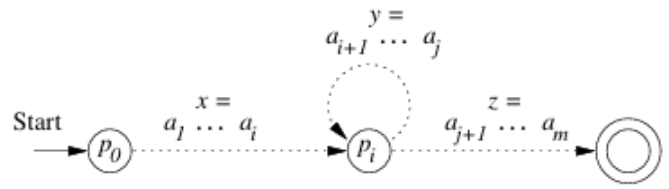


Figure 4.1: Every string longer than the number of states must cause a state to repeat

Pumping Lemma for Regular Languages

Now, consider what happens if the automaton A receives the input xy^kz for any $k \geq 0$. If $k = 0$, then the automaton goes from the start state q_0 (which is also p_0) to p_i on input x. Since p_i is also p_j, it must be that A goes from p_i to the accepting state shown in Fig. 4.1 on input z. Thus, A accepts xz.

If $k > 0$, then A goes from q_0 to p_i on input x, circles from p_i to p_i k times on input y^k, and then goes to the accepting state on input z. Thus, for any $k \geq 0$, xy^kz is also accepted by A; that is, xy^kz is in L. □

- Section 3.2.1 and 3.2.2 of Chapter 3
- Section 4.1 of Chapter 4